IS THE AGENT FRAGILE, OR THE BENCHMARK STATEFUL? AUDITING TASK-ORDER EFFECTS IN SELF-IMPROVING WEB AGENTS

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ABSTRACT

Task-order stress tests are increasingly used to evaluate self-improving agents. The same tasks are permuted while the agent accumulates experience, so the resulting performance change is treated as sensitivity to learning history. We show that stateful web benchmarks turn this experiment into a joint intervention. In WebArena, earlier tasks mutate shared services and can change the ground truth of later tasks. A permutation therefore changes agent memory history and benchmark environment history at the same time. We audit the released trajectories of a recent study that reports task-order fragility for Agent Workflow Memory (AWM) and ReasoningBank. A direct released trajectory shows a purchase reaching checkout despite benchmark score zero. A later task reads the resulting current order correctly and then fails against stale fixed ground truth. Three preregistered dynamic-reference sentinels reproduce this mechanism across both memory methods. All 18 ordinal evaluations pass, while all 36 shuffled evaluations fail. A conservative retrospective over 84 run-domain cells identifies 887 state-exposed reader units in the shuffled memory runs; 424 fail. Removing state-sensitive readers reduces AWM's measured ordinal-to-shuffle gap by 34.6–44.5% and ReasoningBank's by 14.9–17.5%. The remaining gap captures residual order sensitivity after the environment-sensitive stratum is removed. These results establish environment carryover as a material confound in published task-order evaluation. We introduce a state-isolated order-stress protocol that restores service state per task and adds a matched shuffled no-memory control. Live-world evaluations can preserve evolving services through current-state effect scoring. The protocol isolates memory-history sensitivity while retaining interactive benchmark realism.

1 INTRODUCTION

Memory-based agents improve over task streams by storing reusable experience. Agent Workflow Memory (AWM) induces workflows from previous trajectories (Wang et al., 2024). ReasoningBank distills reasoning memories from accumulated experience (Ouyang et al., 2026). A recent reliability study by Ye et al. (2026) tests whether these gains survive changes in task order. Its WebArena experiments compare the default ordinal stream with two random permutations and report substantial degradation under shuffling. The result provides important evidence that self-improvement depends on history.

A clean task-order experiment requires the permutation to act on the learning history under a stable evaluation problem. WebArena introduces a second persistent history because its websites are functional stateful services (Zhou et al., 2024). A successful purchase creates a new order. A cancellation changes the canceled-order set. When tasks share the backend within a run, shuffling changes which service mutations occur before each query. The evaluator can then score a current-state answer against reference text derived from an earlier service state.

This paper identifies the hidden intervention inside self-improving WebArena order tests. Task order jointly perturbs the agent's persistent memory and the benchmark's persistent environment. The measured order gap therefore contains a memory-history component together with an environment-history component. We reconstruct both paths from the released execution code and verify the mechanism in released trajectories. The resulting audit turns a qualitative confound into a measurable source of score change.

The direct witness is decisive. In a released AWM shuffle run, task 509 reaches checkout and creates a new Cole Haan order even though the benchmark scorer assigns zero. Task 96 later opens the current latest order and reads that new object correctly. Its evaluator still expects the earlier canceled order, so the task fails. This same mechanism replicates on three preregistered dynamic-reference sentinels. Across AWM and ReasoningBank, all 18 ordinal method–run units pass. Across the two shuffled streams, all 36 corresponding units fail.

The effect also changes aggregate conclusions. Our released-trajectory retrospective identifies 887 lower-bound state-exposed reader units in shuffled memory runs; 424 fail. Removing preregistered state-sensitive readers reduces AWM's measured shuffle gap by 34.6–44.5%. The corresponding reduction for ReasoningBank is 14.9–17.5%. A residual gap remains after this exclusion, which separates the environment-sensitive component from the remaining order sensitivity.

Recent benchmark work already recognizes the broader challenge of stateful truth. DynamicMCP-Bench uses effect-based scoring on live stateful servers (Kamiński et al., 2026). ClawMark evaluates evolving service state with deterministic post-state checkers (Meng et al., 2026). We introduce a source-grounded causal audit of task-order evaluation in self-improving agents and quantify how persistent benchmark state enters the published measurement. This connection exposes a new requirement for longitudinal agent evaluation: the benchmark's history must become an explicit experimental variable whenever the agent's history is manipulated.

Contributions.

- We formalize task-order evaluation on a stateful benchmark as a joint intervention on memory history and environment history.
- We produce a content-addressed released-trajectory witness that links an upstream backend mutation to a downstream stale-ground-truth failure.
- We establish a replicated sentinel panel with 18/18 ordinal passes and 0/36 shuffled passes across AWM and ReasoningBank.
- We quantify the aggregate measurement effect through a released-trajectory retrospective and a state-sensitivity exclusion analysis.
- We introduce a state-isolated order-stress protocol for clean longitudinal evaluation of self-improving agents.

2 Task Order in a Stateful Evaluation

Let π denote a permutation of benchmark tasks. A memory-based agent maintains persistent state M_t that is updated after task t. A stateful benchmark maintains environment state E_t, such as the order database behind a shopping website. Executing task i with action trajectory a_i induces

$$M_{t+1} = U_M(M_t, x_i, a_i, y_i), \tag{1}$$
$$E_{t+1} = U_E(E_t, a_i), \tag{2}$$

where x_i is the task and y_i is observed feedback. Under order π, the score for task i is

$$Y_i(\pi) = F(M_{<i}(\pi), E_{<i}(\pi), x_i, G_i)\,, \tag{3}$$

where G_i is the evaluator. A memory-history estimand requires the compared orders to preserve the same environment path, $E(\pi_1) \equiv E(\pi_0)$. A live-state evaluator can achieve the same identification goal by deriving G_i from the environment reached at execution time.

Figure 1 exposes the hidden causal path. The order intervention reaches the score through memory history and through environment history. Fixed reference text introduces an additional failure mode when it continues to describe an earlier service state.

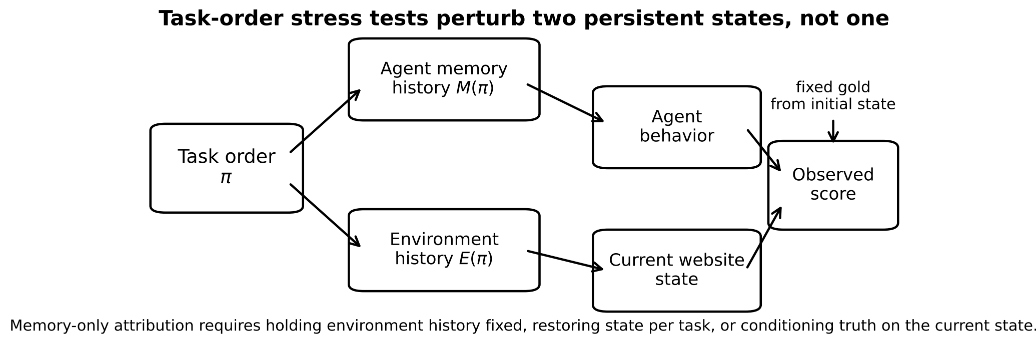

Figure 1: Task order jointly intervenes on the self-improving agent and the stateful benchmark. A clean memory-history estimate controls the environment-history path.

A fresh browser context does not restore a functional service database. The browser can restart while earlier purchases remain in the shopping backend. The relevant experimental state is therefore the persistent service state reached by previous benchmark tasks.

3 Audit Methodology

3.1 Audited system and released evidence

We audit the WebArena portion of Ye et al. (2026). The release evaluates online AWM and ReasoningBank on six WebArena domains. Each method is tested on the ordinal stream and on two shuffled streams with seeds 42 and 99. Three independent runs are released for each condition. Execution is sequential within a domain. The website-reset script runs before the domain-level stream, while the per-task reset reconstructs the browser context without restoring the website backend.

All quantitative analyses use released benchmark configurations and released trajectory metadata. We organize the audit into three evidence levels.

Protocol audit. We inspect the execution path to determine when the service state is restored. We also inspect WebArena task definitions to identify persistent writers and tasks whose truth depends on mutable relational state.

Direct witness. We require a released trajectory in which an upstream state mutation can be linked to the current state observed by a downstream task. We then compare that observed state with the fixed evaluator used by the release.

Exposure retrospective. We define high-confidence writer families and state-sensitive reader families in the four mutable single-site domains. A reader is marked exposed when an earlier benchmark-successful task writes the same preregistered state channel. This rule is fixed before joining with downstream outcomes.

3.2 Dynamic-reference sentinels

We preregister three tasks whose truth is defined by mutable recency. Shopping task 96 asks for the status of the latest order; its fixed reference describes a canceled order. Shopping task 189 asks for the total of the latest pending order; its fixed reference is \$754.99. Shopping Admin task 198 asks for the customer on the most recent canceled order; its fixed reference is Lily Potter. Selection uses task semantics before model outcomes are inspected. We compare each task across AWM and ReasoningBank. Every comparison uses the same three runs under the ordinal stream and both shuffle seeds.

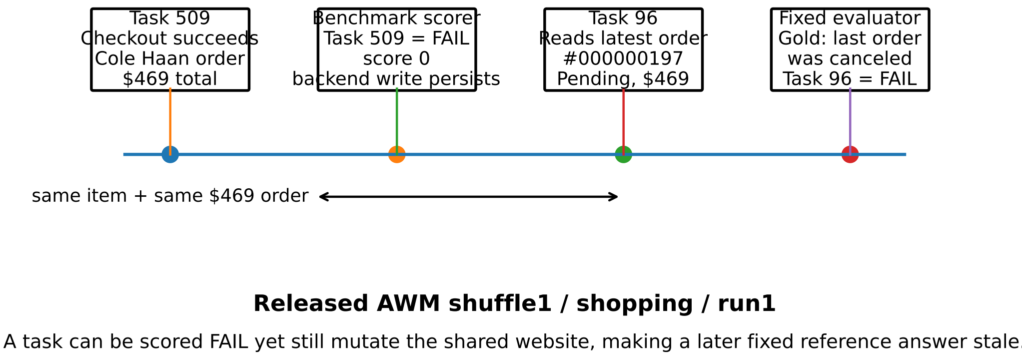

Figure 2: Direct released-trajectory witness. A scorer-failed purchase persists in the shopping backend. The downstream agent observes the resulting current order while the evaluator continues to score an earlier service state.

3.3 Aggregate sensitivity analysis

The static audit identifies 112 high-confidence state-sensitive readers among the 812 WebArena tasks and 244 persistent mutators. Reconstructing the tested streams yields 1,236 direct writer–reader pairs. Of these pairs, 1,005 change exposure ordering across the evaluated streams. We join the frozen task semantics to the released $\{\texttt{task\_id}, \texttt{is\_successful}\}$ columns of each trajectory parquet.

We measure the published order contrast twice. The first estimate uses every released task. The second removes the preregistered state-sensitive readers from every order before recomputing success. The change between these two estimates quantifies how strongly mutable-reference tasks support the measured order gap.

4 A Direct Ground-Truth Invalidation Witness

Figure 2 reconstructs a released AWM Shopping trajectory under shuffle seed 42. Task 509 asks the agent to purchase a Cole Haan dress chukka boot. The agent reaches the checkout success page. The log records a $464 subtotal with $5 shipping, yielding a $469 order. The benchmark scorer assigns FAIL with score 0, yet the completed checkout remains in the shopping backend.

Twenty-six minutes later, task 96 asks for the status of the latest order. The agent opens the account and reads Order #000000197. The order is Pending and contains the same Cole Haan item with the same $469 total. The agent reports this current state. Its evaluator still expects the earlier canceled order, so the task receives FAIL. Under the ordinal stream, task 96 occurs before the purchase tasks and all six AWM/ReasoningBank ordinal replications pass.

The witness establishes that benchmark score and environment mutation are separate state transitions. It also establishes that a current-state answer can be semantically correct while fixed reference text has become stale. This writer–reader chain supplies a direct mechanism for order-dependent evaluation error.

5 Results

5.1 Dynamic-reference failures replicate across methods and orders

Figure 3 reports the three sentinels. Across AWM and ReasoningBank, every ordinal replicate passes: 18/18 method–run units. Across the two shuffled streams, none passes: 0/36. For each sentinel under each shuffle seed, all six paired method–run comparisons move from ordinal PASS to shuffled FAIL. The reverse transition never occurs. The two-sided exact sign test gives $p = 0.03125$ for every six-pair panel.

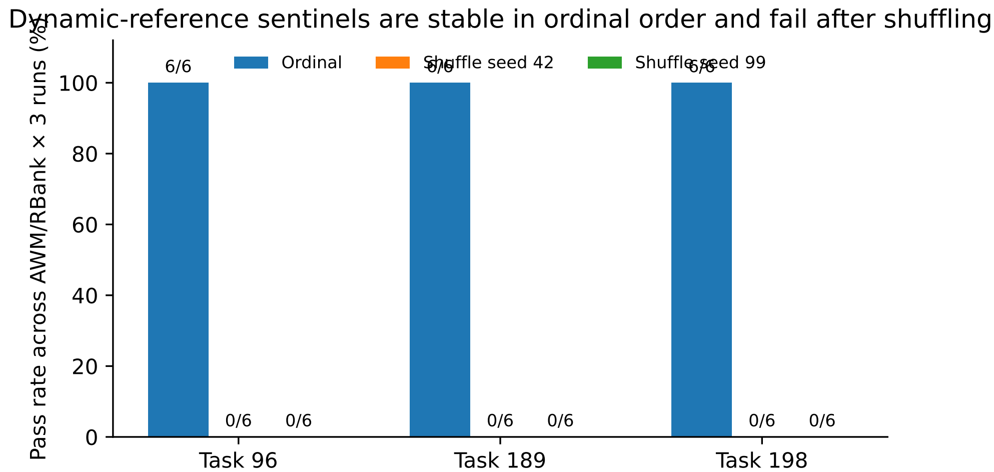

Figure 3: Three dynamic-reference sentinels across AWM and ReasoningBank. Each bar aggregates three runs per method. All 18 ordinal units pass; all 36 shuffled units fail.

Table 1: Released-trajectory retrospective using the frozen writer–reader exposure rule.

Method	Order	Success	Exposed	Exp. fail	Clean fail
AWM	ordinal	56.43%	60	18.33%	33.33%
AWM	seed 42	51.99%	227	49.78%	33.95%
AWM	seed 99	51.30%	208	48.56%	37.50%
ReasoningBank	ordinal	59.73%	58	27.59%	31.30%
ReasoningBank	seed 42	52.12%	224	45.98%	39.29%
ReasoningBank	seed 99	52.32%	228	46.93%	38.89%

Task 189 shows the same ground-truth shift at the value level. One AWM shuffle run identifies Order #000000191 as the current latest pending order with total $478.42. The fixed reference remains $754.99.

Task 198 reproduces the mechanism on Shopping Admin. A shuffled run identifies Sarah Miller on the current most recent canceled order #000000299. The fixed reference remains Lily Potter. Earlier shuffled tasks directly mutate the order database, so the changed reference object follows from the tested task stream itself.

5.2 State exposure is common in released shuffles

Across the 48 shuffled memory-method run-domain cells in the four mutable single-site domains, the strict exposure detector identifies 887 state-sensitive reader units that follow at least one overlapping benchmark-successful writer. Of these units, 424 fail. Table 1 shows the corresponding failure rates. Exposure becomes substantially more common under shuffling because mutating tasks move ahead of dynamic-reference queries.

For AWM, exposed-reader failure reaches 49.78% under seed 42 and 48.56% under seed 99. The corresponding clean-reader rates are 33.95% and 37.50%. ReasoningBank shows the same direction: exposed-reader failure reaches 45.98% and 46.93%, compared with 39.29% and 38.89% for clean readers. The retrospective therefore connects the direct witness to a broader population of order-sensitive measurement objects.

5.3 State-sensitive tasks account for a material part of the order gap

Figure 4 compares the released ordinal-to-shuffle success gap with the same contrast after state-sensitive readers are excluded. For AWM, the raw gap is 4.44 percentage points under seed 42 and

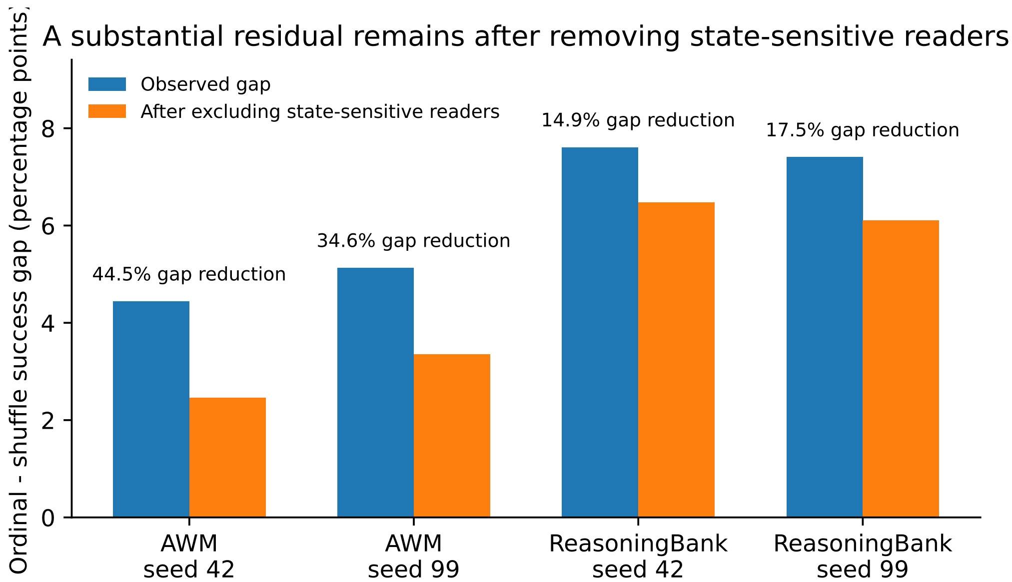

Figure 4: Observed task-order gap before and after excluding preregistered state-sensitive readers. Mutable-reference tasks account for a material fraction of the measured gap, especially for AWM.

5.13 points under seed 99. The filtered gaps are 2.46 and 3.36 points, corresponding to reductions of 44.5% and 34.6%.

ReasoningBank starts with raw gaps of 7.61 and 7.40 points. After the same exclusion, the gaps are 6.48 and 6.11 points. The reductions are 14.9% and 17.5%.

The released task-order effect therefore contains an identifiable environment-sensitive component. After state-sensitive readers are removed, AWM retains 2.46–3.36 percentage points of order sensitivity. ReasoningBank retains more than six percentage points. The residual defines the effect that a state-isolated rerun can attribute more directly to self-improvement history.

6 A State-Isolated Order-Stress Protocol

Longitudinal self-improvement evaluation should intervene on one persistent history at a time. We therefore define a state-isolated order-stress protocol around the causal graph in Figure 1.

Treat order as the learning-history intervention. The ordinal stream and each shuffled stream define the experience history presented to the agent. Corresponding task evaluations use the same preregistered service state so that the manipulated variable remains the agent's history.

Restore environment state per task. Each task starts from a content-addressed service snapshot. Task-specific service instances provide an equivalent implementation. This blocks the $\pi \to E \to Y$ path while preserving $\pi \to M \to Y$.

Use current-state scoring for intentionally live worlds. A live benchmark can derive truth from the service state reached at execution time. Effect-based evaluators provide the same property by checking the requested state transition. DynamicMCPBench implements effect checkpoints for stateful servers (Kamiński et al., 2026). ClawMark uses deterministic post-state checks in evolving sandbox services (Meng et al., 2026). Under this design, a phrase such as “latest order” is resolved against the current order relation.

Shuffle the no-memory control. A complete order-stress experiment evaluates the same agent scaffold without persistent memory under every tested order. Define

$$\Delta_{\mathrm{mem}} = S_{\mathrm{mem,ord}} - S_{\mathrm{mem,shuf}}, \tag{4}$$

$$\Delta_{\mathrm{base}} = S_{\mathrm{base,ord}} - S_{\mathrm{base,shuf}}. \tag{5}$$

The difference $\Delta_{\mathrm{mem}} - \Delta_{\mathrm{base}}$ estimates the incremental order sensitivity introduced by persistent self-improvement under the matched execution scaffold.

Report state-sensitive strata. Benchmark authors should preregister tasks whose targets are defined by mutable relational state. Reporting this stratum separately makes movement in the measurement object visible inside the final order-stress result.

7 Related Work

Self-improving agent memory. AWM learns reusable workflows from prior trajectories and supports online memory induction (Wang et al., 2024). ReasoningBank stores distilled reasoning strategies from accumulated experience and uses memory-aware test-time scaling (Ouyang et al., 2026). The fragility study we audit advances this line by explicitly testing run-to-run variance and task order (Ye et al., 2026). Our audit identifies the benchmark-state path activated by the same order intervention.

Stateful agent benchmarks. WebArena provides realistic functional web applications with programmatic task evaluators (Zhou et al., 2024). DynamicMCPBench makes service evolution a first-class evaluation variable and motivates effect-based scoring on live servers (Kamiński et al., 2026). ClawMark evaluates long-running agents in evolving sandbox services through deterministic checks over post-execution state (Meng et al., 2026). Our work links state-aware truth to longitudinal self-improvement evaluation by showing that task-order interventions also transform benchmark history.

8 Limitations

The released no-memory baseline uses the ordinal stream only. A matched shuffled no-memory run remains experiment debt. A per-task-restored rerun is the second prospective control. Until those runs are added, the 34.6–44.5% and 14.9–17.5% values remain task-exclusion sensitivity reductions. They do not constitute a causal mediation percentage for the full published gap.

The exposure detector uses a frozen rule set over explicit state channels. It omits mutations from scorer-failed writers when computing the 887-unit aggregate. It can also miss hidden service dependencies. The direct task-509 witness proves that the aggregate is a lower bound. An exposure label alone does not establish that a particular writer changed the exact answer required by a reader; the sentinel panel supplies that higher-precision evidence for three dynamic-reference tasks.

The empirical audit covers one WebArena release from one self-improvement study. The identification problem applies to stateful evaluation in general, while the reported prevalence values remain specific to this release. Multi-benchmark replication is therefore experiment debt for a broader empirical claim.

A substantial residual order gap remains after state-sensitive readers are excluded. Environment carryover therefore cannot account for the entire observed order effect. The paper's main empirical claim is material contamination of task-order measurement, with residual self-improvement sensitivity preserved as an active phenomenon.

The prospective contrast $\Delta_{\mathrm{mem}} - \Delta_{\mathrm{base}}$ assumes that the no-memory scaffold captures the environment-order contribution under the same execution stack. A memory method may interact with mutated state in ways the baseline does not reproduce. A factorial design with per-task restoration can resolve that interaction directly.

9 Conclusion

Task-order stress testing on a stateful benchmark is a joint intervention. The permutation changes the agent's experience history and the benchmark's service history. Released AWM trajectories contain a direct writer–reader ground-truth invalidation chain. The same mechanism reproduces perfectly across three dynamic-reference sentinels. Aggregate state-sensitive exclusion removes a material fraction of the reported order gap, especially for AWM.

These findings establish benchmark history as a first-class variable in longitudinal agent evaluation. The state-isolated order-stress protocol restores the intended scientific intervention by fixing service state while task order changes. Live-world variants can condition scoring on current service state. With benchmark history made explicit, task-order experiments can measure self-improvement history directly and support stronger claims about long-horizon agent evolution.

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A Evidence and Reproduction Details

A.1 Frozen evidence artifacts

The audit is generated from four machine-readable artifact classes. The task-interference audit covers all 812 official WebArena task configurations. The direct ground-truth invalidation certificate binds the writer–reader witness to released logs. The sentinel panel stores every method–run outcome for the three dynamic-reference tasks. The released-trajectory retrospective stores the 84 run-domain cells used for aggregate analysis. Raw author logs and trajectory metadata are cached by content digest outside the manuscript source tree. Headline numbers are compiled from these artifacts.

A.2 Sentinel definitions

Task 96 asks for the status of the latest Shopping order. Task 189 asks for the total of the latest pending Shopping order. Task 198 asks for the customer associated with the most recent canceled

Shopping Admin order. Each target is defined by a mutable relation. The criterion is frozen from task semantics before downstream outcomes are joined.

A.3 EXPOSURE RULE

A reader is marked exposed when its semantics depends on a preregistered mutable state channel. An earlier task in the tested stream must write the same channel. That writer must receive benchmark success. This rule produces the exposure counts used in Table 1.

A.4 CLAIM LEDGER

The manuscript is bound to a frozen claim ledger. C1 records the joint memory-history and environment-history intervention. C2 records the released writer–reader invalidation witness. C3 records the replicated sentinel effect. C4 records the aggregate state-sensitivity result. C5 records the residual order sensitivity that remains after the state-sensitive exclusion. Every headline statement in the main paper maps to one of these evidence-bound claims.

A.5 FULL SENSITIVITY CONTRASTS

Table 2: Raw and state-sensitive-excluded ordinal-to-shuffle success gaps. Values are percentage points. "Removed" is the reduction fraction of the raw gap.

Method	Shuffle	Raw gap	Excluded gap	Raw gap removed
AWM	seed 42	4.44	2.46	44.5%
AWM	seed 99	5.13	3.36	34.6%
ReasoningBank	seed 42	7.61	6.48	14.9%
ReasoningBank	seed 99	7.40	6.11	17.5%